TASK 2

a) **Ingest structured data using Apache Sqoop.**

Verify Sqoop

sqoop version

Start the MySQL Server

sudo service mysql start

log in to MySQL

mysql -u root -p

pwd:cloudera

create database

CREATE DATABASE company;

USE company;

Create table:

```
CREATE TABLE employees (
   id INT PRIMARY KEY,
   name VARCHAR(50),
   age INT,
   salary FLOAT
);
```

Insert records

```
INSERT INTO employees VALUES
(1, 'John', 30, 50000),
(2, 'Alice', 28, 60000),
(3, 'Bob', 35, 75000);
```

Retrieve records:

SELECT * FROM employees;

Exit MySQL:

exit;

List all databases in MySQL using sqoop:

sqoop list-databases --connect jdbc:mysql://localhost:3306/ --username root --password cloudera

list tables inside the company database:

sqoop list-tables --connect jdbc:mysql://localhost:3306/company --username root --password cloudera

import operation:

sqoop import --connect jdbc:mysql://localhost:3306/vb --username root --password cloudera --table emp --m 1 --target-dir gh

```
[cloudera@quickstart Desktop]$ sqoop import --connect jdbc:mysql://localhost:330
6/vb --username root --password cloudera --table emp --m 1 --target-dir gh
```

```
                Total time spent by all reduces in occupied slots (ms)=0
                Total time spent by all map tasks (ms)=4088
                Total vcore-milliseconds taken by all map tasks=4088
                Total megabyte-milliseconds taken by all map tasks=4186112
        Map-Reduce Framework
                Map input records=4
                Map output records=4
                Input split bytes=87
                Spilled Records=0
                Failed Shuffles=0
                Merged Map outputs=0
                GC time elapsed (ms)=91
                CPU time spent (ms)=1320
                Physical memory (bytes) snapshot=224751616
                Virtual memory (bytes) snapshot=1593380864
                Total committed heap usage (bytes)=278396928
        File Input Format Counters
                Bytes Read=0
        File Output Format Counters
                Bytes Written=55
25/03/24 01:54:01 INFO mapreduce.ImportJobBase: Transferred 55 bytes in 22.5721
seconds (2.4366 bytes/sec)
25/03/24 01:54:01 INFO mapreduce.ImportJobBase: Retrieved 4 records.
[cloudera@quickstart Desktop]$
```

check the data:

hdfs dfs -cat gh/*

```
[cloudera@quickstart Desktop]$ hdfs dfs -ls gh
Found 2 items
-rw-r--r--   1 cloudera cloudera          0 2025-03-24 01:53 gh/_SUCCESS
-rw-r--r--   1 cloudera cloudera         55 2025-03-24 01:53 gh/part-m-00000
[cloudera@quickstart Desktop]$ hdfs dfs -cat gh/*
201,abc,5000
202,abcd,8000
203,bcd,15000
204,bacd,9000
[cloudera@quickstart Desktop]$
```

export operation:

sqoop export --connect jdbc:mysql://localhost:3306/vb --username root --password cloudera --table emp --export-dir emp/* --input-fields-terminated-by ',' --columns "id,name,age" --m 1

Output:

```
[cloudera@quickstart Desktop]$ sqoop export --connect jdbc:mysql:///localhost:330
6/vb --username root --password cloudera --table emp --export-dir gh/* --input-f
ields-terminated-by ',' --columns "eid,ename,sal" --m 1
```

```
                Total vcore-milliseconds taken by all map tasks=3725
                Total megabyte-milliseconds taken by all map tasks=3814400
        Map-Reduce Framework
                Map input records=4
                Map output records=4
                Input split bytes=142
                Spilled Records=0
                Failed Shuffles=0
                Merged Map outputs=0
                GC time elapsed (ms)=86
                CPU time spent (ms)=1140
                Physical memory (bytes) snapshot=221257728
                Virtual memory (bytes) snapshot=1589104640
                Total committed heap usage (bytes)=278396928
        File Input Format Counters
                Bytes Read=0
        File Output Format Counters
                Bytes Written=0
25/03/24 01:57:17 INFO mapreduce.ExportJobBase: Transferred 200 bytes in 15.7463
 seconds (12.7014 bytes/sec)
25/03/24 01:57:17 INFO mapreduce.ExportJobBase: Exported 4 records.
[cloudera@quickstart Desktop]$
```

Before:

```
mysql> select * from emp;
+------+-------+-------+
| eid  | ename | sal   |
+------+-------+-------+
|  201 | abc   |  5000 |
|  202 | abcd  |  8000 |
|  203 | bcd   | 15000 |
|  204 | bacd  |  9000 |
+------+-------+-------+
4 rows in set (0.00 sec)
```

After:

```
mysql> select * from emp;
+------+-------+-------+
| eid  | ename | sal   |
+------+-------+-------+
|  201 | abc   |  5000 |
|  202 | abcd  |  8000 |
|  203 | bcd   | 15000 |
|  204 | bacd  |  9000 |
|  201 | abc   |  5000 |
|  202 | abcd  |  8000 |
|  203 | bcd   | 15000 |
|  204 | bacd  |  9000 |
+------+-------+-------+
8 rows in set (0.00 sec)
```

B) Ingest structured and unstructured data using Flume.

Structured Data (MySQL Data to HDFS)

1. verify Flume is installed: flume-ng version
2. Configure Structured Data: sudo nano /etc/flume-ng/conf/mysql-flume.conf

   agent1.sources = mysql_source
   agent1.sinks = hdfs_sink
   agent1.channels = mem_channel
   agent1.sources.mysql_source.type = exec
   agent1.sources.mysql_source.command = mysql -u root -p'yourpassword' -e "SELECT * FROM database.table;"
   agent1.sources.mysql_source.shell = /bin/bash -c
   agent1.channels.mem_channel.type = memory
   agent1.channels.mem_channel.capacity = 1000
   agent1.channels.mem_channel.transactionCapacity = 100
   agent1.sinks.hdfs_sink.type = hdfs
   agent1.sinks.hdfs_sink.hdfs.path = hdfs://localhost:8020/user/flume/mysql_data/
   agent1.sinks.hdfs_sink.hdfs.fileType = DataStream
   agent1.sinks.hdfs_sink.hdfs.writeFormat = Text
   agent1.sinks.hdfs_sink.hdfs.rollInterval = 10
   agent1.sources.mysql_source.channels = mem_channel
   agent1.sinks.hdfs_sink.channel = mem_channel
3. Start the Flume Agents: flume-ng agent --conf /etc/flume-ng/conf/ --name agent1 --conf-file /etc/flume-ng/conf/mysql-flume.conf -Dflume.root.logger=INFO,console
4. Verify Data in HDFS: hdfs dfs -ls /user/flume/mysql_data/

Output:

```
[cloudera@quickstart Desktop]$ nano mysql-flume.conf
[cloudera@quickstart Desktop]$ flume-ng agent --conf /home/cloudera/Desktop/ --n
ame agent1 --conf-file /home/cloudera/Desktop/mysql-flume.conf -Dflume.root.logg
er=INFO,console
Info: Including Hadoop libraries found via (/usr/bin/hadoop) for HDFS access
Info: Including HBASE libraries found via (/usr/bin/hbase) for HBASE access
Info: Including Hive libraries found via () for Hive access
+ exec /usr/java/jdk1.7.0_67-cloudera/bin/java -Xmx20m -Dflume.root.logger=INFO,
console -cp '/home/cloudera/Desktop:/usr/lib/flume-ng/lib/*:/etc/hadoop/conf:/us
r/lib/hadoop/lib/*:/usr/lib/hadoop/.//*:/usr/lib/hadoop-hdfs/./:/usr/lib/hadoop-
hdfs/lib/*:/usr/lib/hadoop-hdfs/.//*:/usr/lib/hadoop-yarn/lib/*:/usr/lib/hadoop-
yarn/.//*:/usr/lib/hadoop-mapreduce/lib/*:/usr/lib/hadoop-mapreduce/.//*:/usr/li
b/hbase/bin/../conf:/usr/java/jdk1.7.0_67-cloudera/lib/tools.jar:/usr/lib/hbase/
bin/..:/usr/lib/hbase/bin/../lib/activation-1.1.jar:/usr/lib/hbase/bin/../lib/ap
acheds-i18n-2.0.0-M15.jar:/usr/lib/hbase/bin/../lib/apacheds-kerberos-codec-2.0.
0-M15.jar:/usr/lib/hbase/bin/../lib/api-asn1-api-1.0.0-M20.jar:/usr/lib/hbase/bi
n/../lib/api-util-1.0.0-M20.jar:/usr/lib/hbase/bin/../lib/asm-3.2.jar:/usr/lib/h
base/bin/../lib/avro.jar:/usr/lib/hbase/bin/../lib/aws-java-sdk-bundle-1.11.134.
jar:/usr/lib/hbase/bin/../lib/commons-beanutils-1.9.2.jar:/usr/lib/hbase/bin/../
lib/commons-beanutils-core-1.8.0.jar:/usr/lib/hbase/bin/../lib/commons-cli-1.2.j
ar:/usr/lib/hbase/bin/../lib/commons-codec-1.9.jar:/usr/lib/hbase/bin/../lib/com
mons-collections-3.2.2.jar:/usr/lib/hbase/bin/../lib/commons-compress-1.4.1.jar:
/usr/lib/hbase/bin/../lib/commons-configuration-1.6.jar:/usr/lib/hbase/bin/../li
b/commons-daemon-1.0.13.jar:/usr/lib/hbase/bin/../lib/commons-digester-1.8.jar:/
25/03/24 01:49:44 INFO instrumentation.MonitoredCounterGroup: Monitored counter
group for type: SOURCE, name: mysql_source: Successfully registered new MBean.
25/03/24 01:49:44 INFO instrumentation.MonitoredCounterGroup: Component type: SO
URCE, name: mysql_source started
25/03/24 01:49:44 INFO instrumentation.MonitoredCounterGroup: Monitored counter
group for type: SINK, name: hdfs_sink: Successfully registered new MBean.
25/03/24 01:49:44 INFO instrumentation.MonitoredCounterGroup: Component type: SI
NK, name: hdfs_sink started
25/03/24 01:49:44 INFO source.ExecSource: Command [mysql -u root -p'cloudera' -e
 "SELECT * FROM vb.emp;"] exited with 0
25/03/24 01:49:44 INFO hdfs.HDFSDataStream: Serializer = TEXT, UseRawLocalFileSy
stem = false
25/03/24 01:49:44 INFO hdfs.BucketWriter: Creating hdfs://localhost:8020/bc/Flum
eData.1742806184311.tmp
25/03/24 01:49:56 INFO hdfs.BucketWriter: Closing hdfs://localhost:8020/bc/Flume
Data.1742806184311.tmp
25/03/24 01:49:56 INFO hdfs.BucketWriter: Renaming hdfs://localhost:8020/bc/Flum
eData.1742806184311.tmp to hdfs://localhost:8020/bc/FlumeData.1742806184311
25/03/24 01:49:56 INFO hdfs.HDFSEventSink: Writer callback called.
^Z
[4]+  Stopped                 flume-ng agent --conf /home/cloudera/Desktop/ --na
me agent1 --conf-file /home/cloudera/Desktop/mysql-flume.conf -Dflume.root.logge
r=INFO,console
```

```
[cloudera@quickstart Desktop]$ hdfs dfs -ls /bc
Found 1 items
-rw-r--r--   1 cloudera supergroup         69 2025-03-24 01:49 /bc/FlumeData.174
2806184311
[cloudera@quickstart Desktop]$ hdfs dfs -cat /bc/*
eid     ename   sal
201     abc     5000
202     abcd    8000
203     bcd     15000
204     bacd    9000
[cloudera@quickstart Desktop]$
```

Unstructured Data (Log Files to HDFS)

1. Configure Unstructured Data: sudo nano /etc/flume-ng/conf/logs-flume.conf

   agent2.sources = log_source
   agent2.sinks = hdfs_sink
   agent2.channels = file_channel
   agent2.sources.log_source.type = exec
   agent2.sources.log_source.command = tail -F /var/log/syslog
   agent2.channels.file_channel.type = file
   agent2.channels.file_channel.capacity = 1000
   agent2.channels.file_channel.transactionCapacity = 100
   agent2.sinks.hdfs_sink.type = hdfs
   agent2.sinks.hdfs_sink.hdfs.path = hdfs://localhost:8020/user/flume/log_data/
   agent2.sinks.hdfs_sink.hdfs.fileType = DataStream
   agent2.sinks.hdfs_sink.hdfs.writeFormat = Text
   agent2.sinks.hdfs_sink.hdfs.rollInterval = 10
   agent2.sources.log_source.channels = file_channel
   agent2.sinks.hdfs_sink.channel = file_channel
2. Start the Flume Agent: flume-ng agent --conf /etc/flume-ng/conf/ --name agent2 --conf-file /etc/flume-ng/conf/logs-flume.conf -Dflume.root.logger=INFO,console
3. Verify Data in HDFS : hdfs dfs -ls /user/flume/log_data/

Send data through:
echo "Hello AIML-C Flume Program " | sudo tee -a /var/log/syslog

Output:

```
[cloudera@quickstart Desktop]$ flume-ng agent --conf /home/cloudera/Desktop/ --n
ame agent2 --conf-file /home/cloudera/Desktop/logs-flume.conf -Dflume.root.logge
r=INFO,console
```

```
25/03/24 01:44:39 INFO file.FileChannel: Queue Size after replay: 0 [channel=fil
e_channel]
25/03/24 01:44:39 INFO node.Application: Starting Sink hdfs_sink
25/03/24 01:44:39 INFO node.Application: Starting Source log_source
25/03/24 01:44:39 INFO source.ExecSource: Exec source starting with command: tai
l -F /var/log/syslog
25/03/24 01:44:39 INFO instrumentation.MonitoredCounterGroup: Monitored counter
group for type: SOURCE, name: log_source: Successfully registered new MBean.
25/03/24 01:44:39 INFO instrumentation.MonitoredCounterGroup: Component type: SO
URCE, name: log_source started
25/03/24 01:44:39 INFO instrumentation.MonitoredCounterGroup: Monitored counter
group for type: SINK, name: hdfs_sink: Successfully registered new MBean.
25/03/24 01:44:39 INFO instrumentation.MonitoredCounterGroup: Component type: SI
NK, name: hdfs_sink started
25/03/24 01:44:45 INFO hdfs.HDFSDataStream: Serializer = TEXT, UseRawLocalFileSy
stem = false
25/03/24 01:44:45 INFO hdfs.BucketWriter: Creating hdfs://localhost:8020/vv/Flum
eData.1742805885466.tmp
25/03/24 01:44:57 INFO hdfs.BucketWriter: Closing hdfs://localhost:8020/vv/Flume
Data.1742805885466.tmp
25/03/24 01:44:57 INFO hdfs.BucketWriter: Renaming hdfs://localhost:8020/vv/Flum
eData.1742805885466.tmp to hdfs://localhost:8020/vv/FlumeData.1742805885466
25/03/24 01:44:57 INFO hdfs.HDFSEventSink: Writer callback called.
```

```
[cloudera@quickstart Desktop]$ echo "Hello Hyderabad " | sudo tee -a /var/log/sy
slog
Hello Hyderabad
```

```
[cloudera@quickstart Desktop]$ hdfs dfs -ls /vv
Found 5 items
-rw-r--r--   1 cloudera supergroup        186 2025-03-24 01:34 /vv/FlumeData.174
2805241779
-rw-r--r--   1 cloudera supergroup         17 2025-03-24 01:34 /vv/FlumeData.174
2805283880
-rw-r--r--   1 cloudera supergroup        185 2025-03-24 01:36 /vv/FlumeData.174
2805357336
-rw-r--r--   1 cloudera supergroup         21 2025-03-24 01:36 /vv/FlumeData.174
2805389315
-rw-r--r--   1 cloudera supergroup         16 2025-03-24 01:36 /vv/FlumeData.174
2805399427
[cloudera@quickstart Desktop]$ hdfs dfs -cat /vv/*
i am goin g sleep
I am teaching while sleeping
I am eating while playing
Hello GRIET
Hello AIML-C Flume Program Task 2
Today is Monday
Hello Bharath
Hello Raj
Hello Narender
Hey GRIET
Hello Hyderabad
```